Research Note: Short-Horizon BTC Direction vs Polymarket 5m Markets

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Status: portfolio research write-up
Stack: Python, LightGBM, Transformer, Binance OHLCV, Polymarket CLOB
Code: this repository

Abstract

I built short-horizon BTC direction models and tried to trade them on Polymarket Up/Down markets. On candle / futures-style evaluation the models looked usable. On Polymarket, after **executable ask prices**, the same signal was typically **negative EV**. Early live PnL (~$200) is treated as noise, not proof of alpha.

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1. Problem

Polymarket runs recurring BTC Up/Down binaries. A share bought at ask (a) pays 1 if you are right and 0 if wrong, so roughly:

[WR - a]

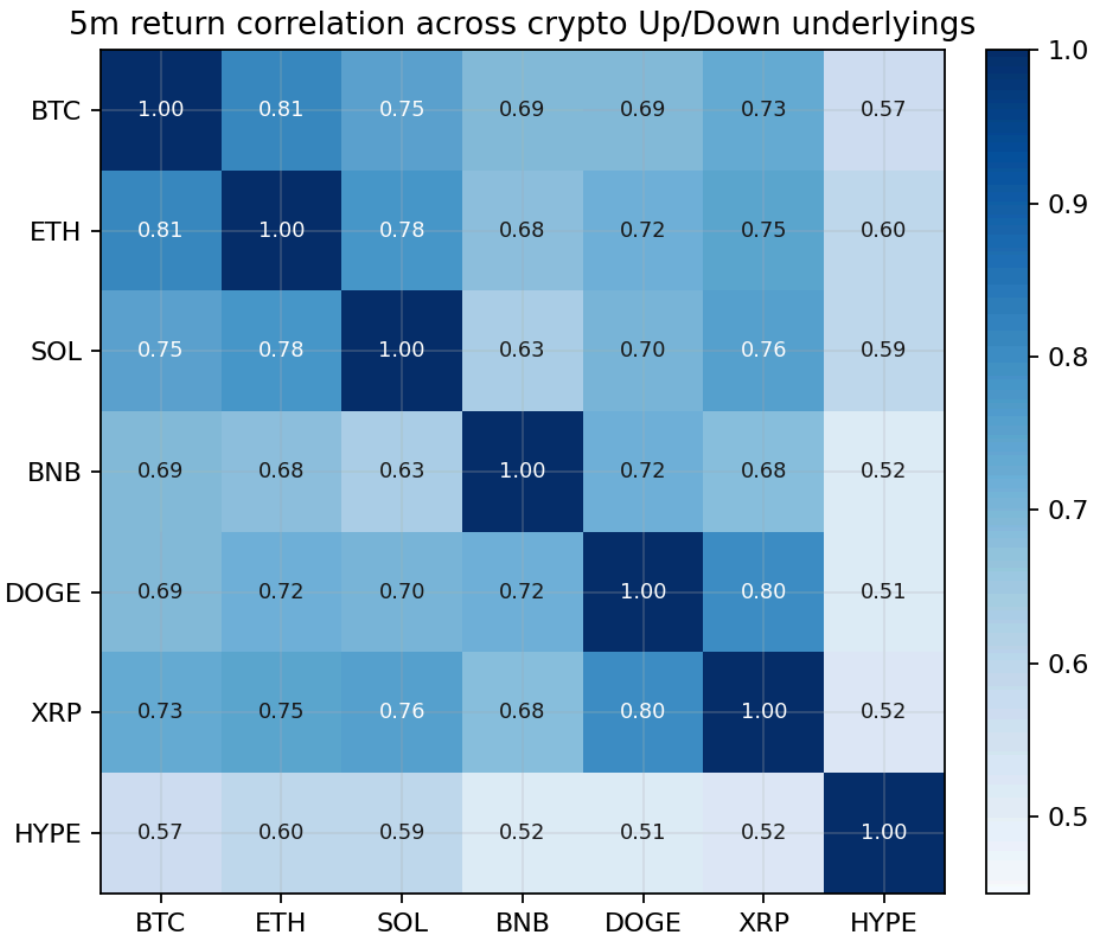
Directional accuracy alone is not enough. You need win rate above the ask you actually pay.

Question: can a Binance-based 5-minute direction model clear that bar on Polymarket?

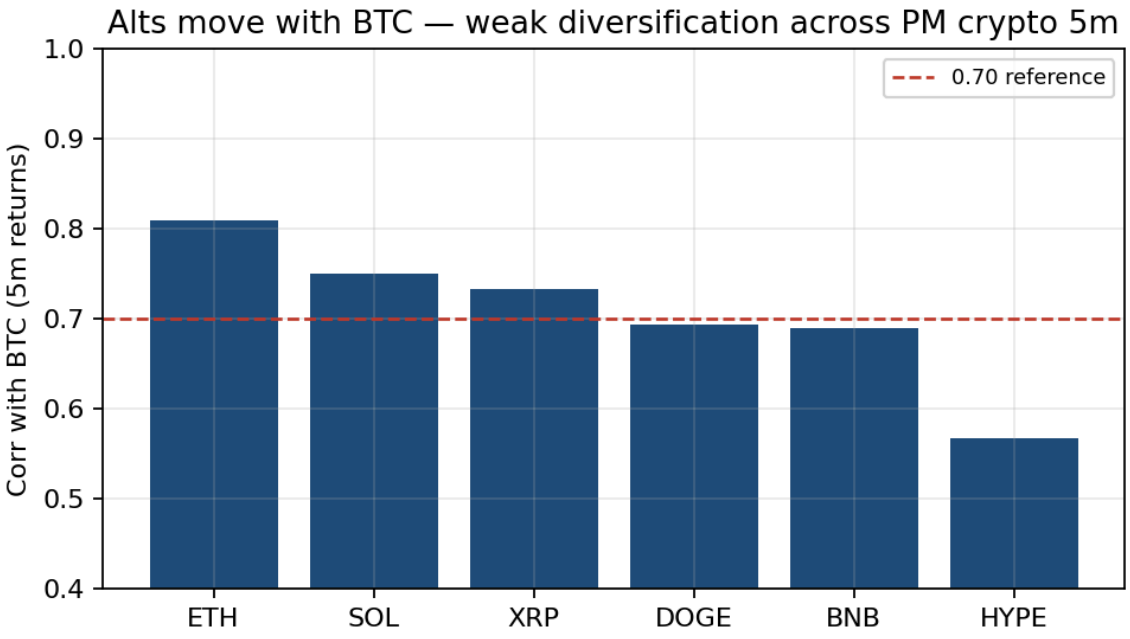
Why BTC (honest), and what about the horizon

Before locking the mainline I scanned **other crypto Up/Down underlyings** on the same market family (ETH, SOL, BNB, DOGE, XRP, HYPE) and also compared **BTC 5m vs 15m** walk-forwards.

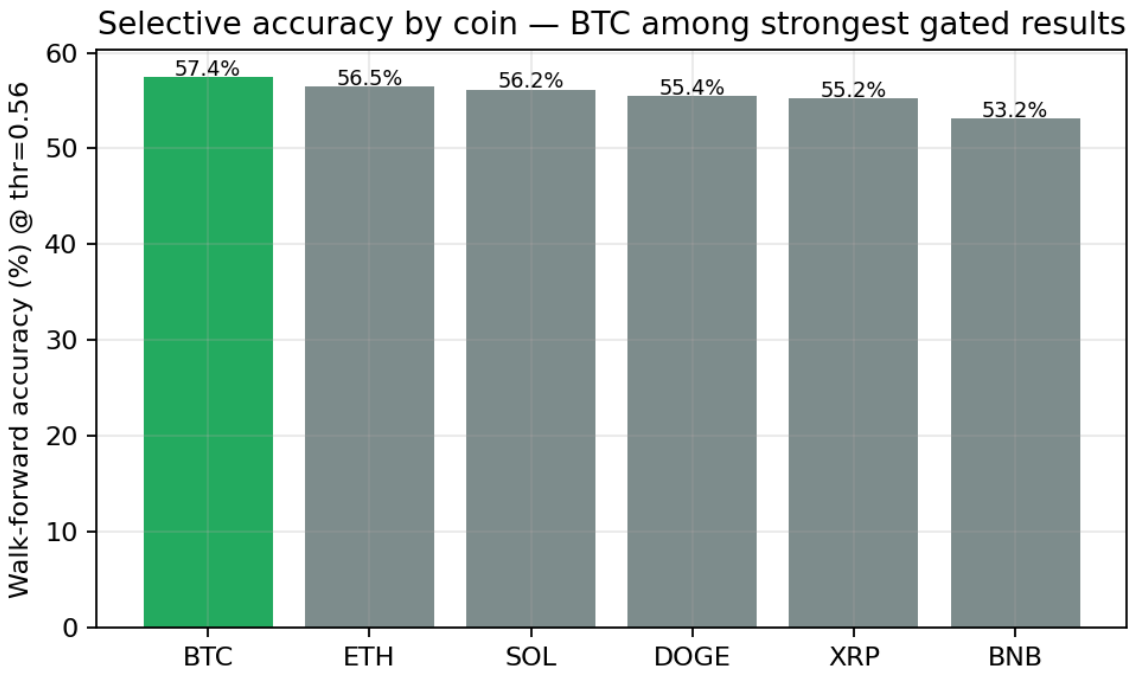
Coin choice — can be stated cleanly.
5m returns are tightly tied to BTC (ETH≈0.81, SOL≈0.75, …); direction agreement with BTC is high (~0.69–0.80). A multi-coin book is mostly **correlated BTC risk**, not diversification — weak reason to merge alts into one "united" strategy. On a common LightGBM walk-forward scan, gated **BTC** was among the strongest coins on selective accuracy, and BTC had the cleanest YES/NO pair availability in a microstructure check of recent crypto 5m markets. So concentrating on **BTC** is justified by the data.



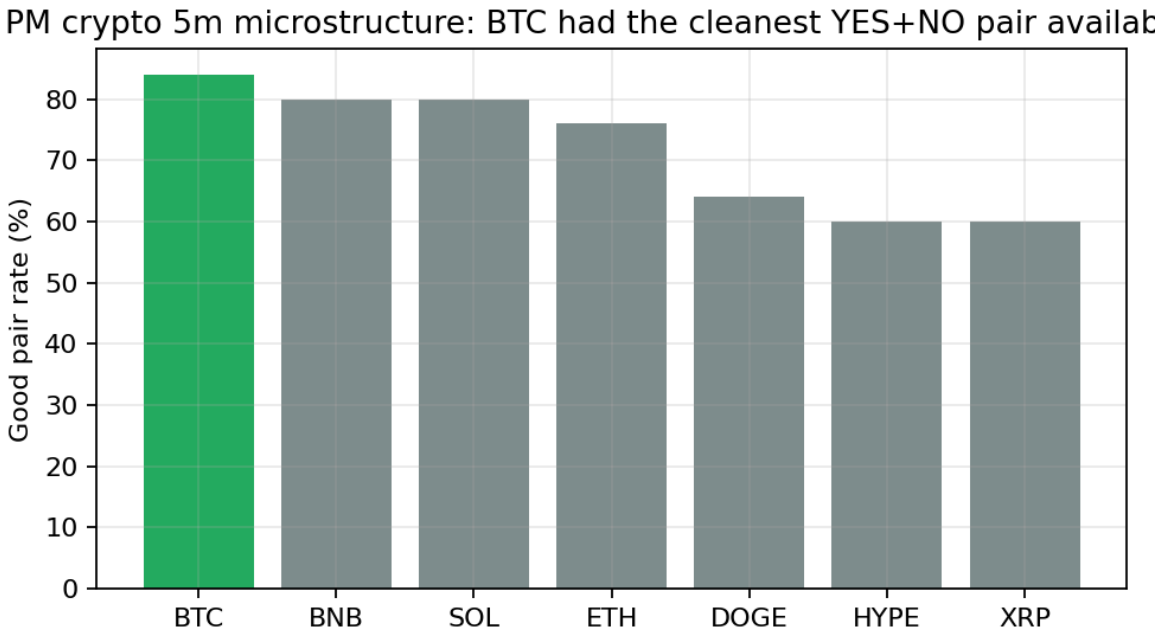
Return correlation heatmap



Correlation with BTC

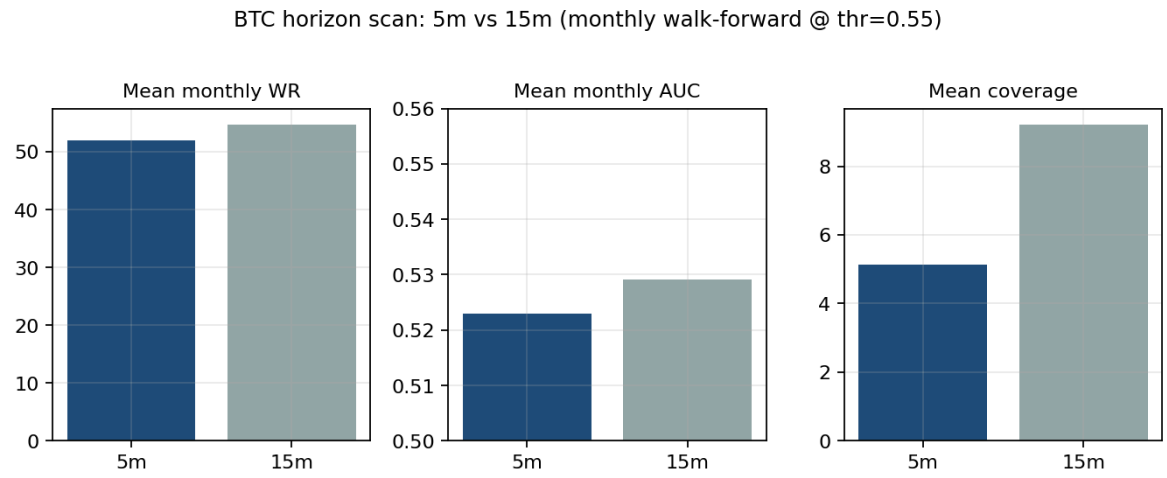


Selective accuracy by coin



Pair quality by coin

Horizon — do not overclaim.
The same lab scan does **not** show that 5m dominated 15m. In the monthly walk-forward file at thr=0.55, **15m looked at least as strong** (higher mean WR / AUC / coverage in that aggregate). This note still centers on **BTC 5m** because that is the system that was built out and live-tested most heavily — not because a horizon bake-off crowned 5m the winner.



5m vs 15m scan

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2. Two evaluations

I kept two scoreboards separate:

1. **Candle evaluation** — predict the next BTC 5m candle; optional futures-style simulator with fees / leverage.
2. **Polymarket evaluation** — same idea mapped onto contract prices, EV at ask, shadow / live fills.

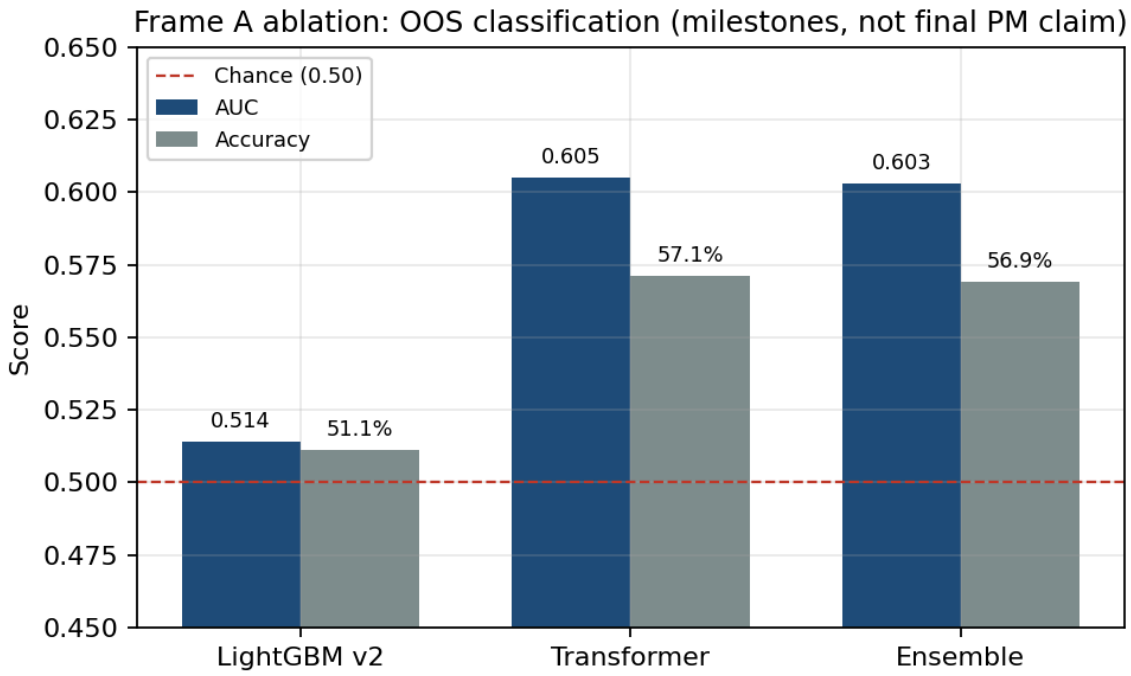
A good candle score does not imply a good Polymarket strategy.

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3. Candle-side work

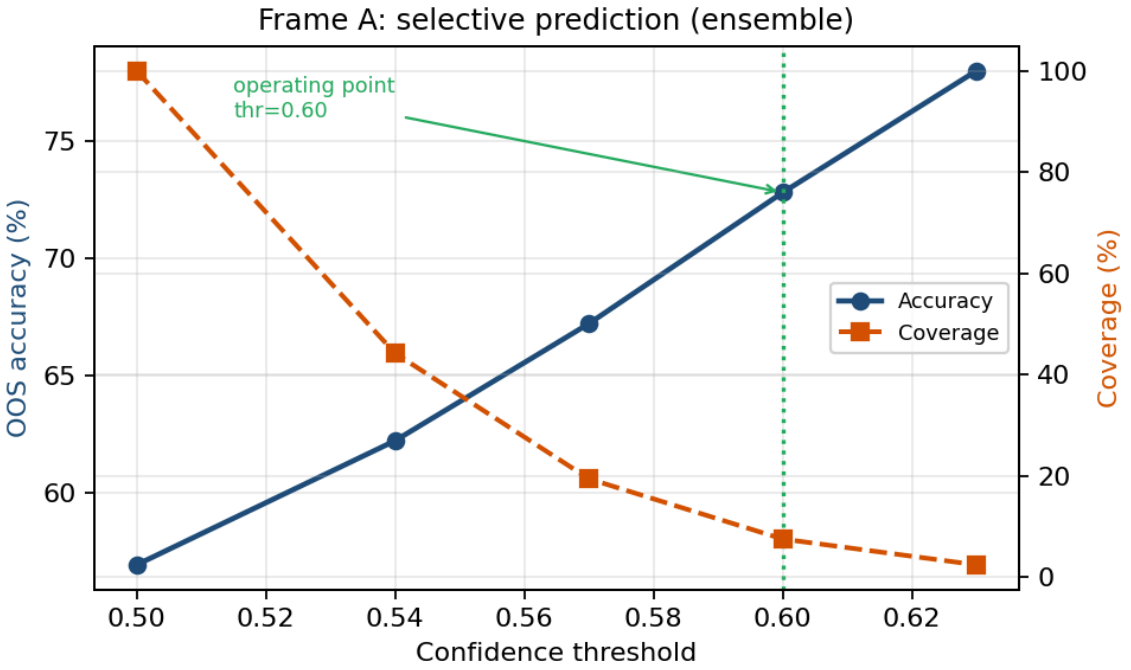
Details: `model_card.md`.

I trained LightGBM and a small candle Transformer on Binance BTC 5m features (ETH used as context), then combined them as a weighted ensemble for experiments. On the held-out candle slice the Transformer dominated discrimination; LightGBM was weak alone; the blend was close to the Transformer.



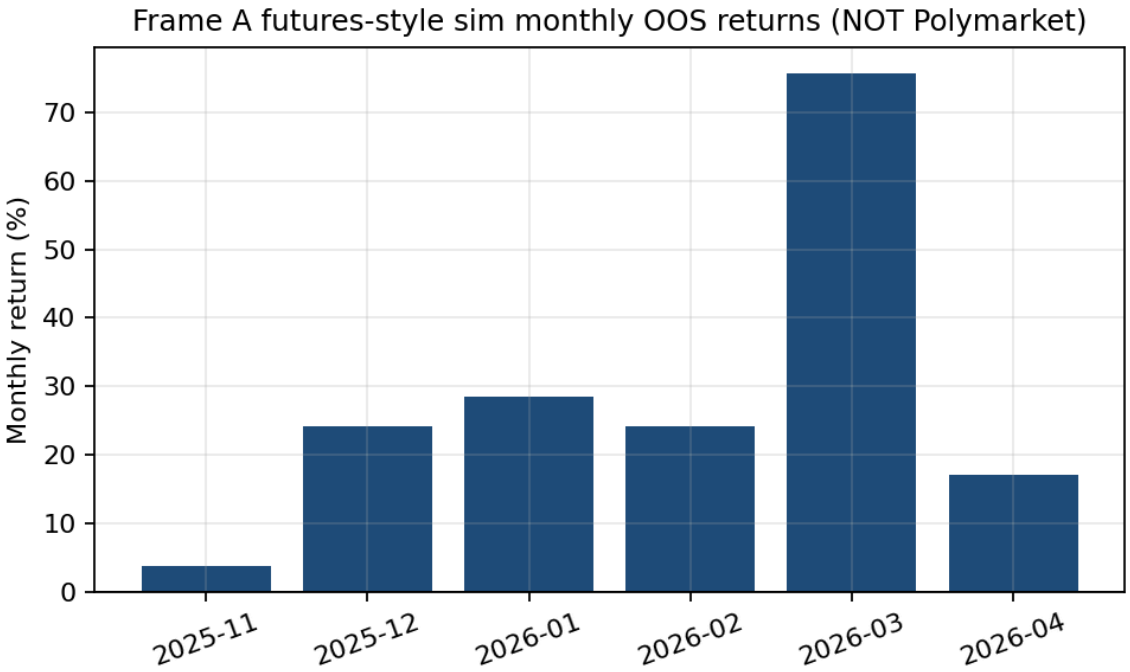
OOS candle metrics

Confidence gating helped: trading only high-confidence bars raised accuracy while cutting coverage.



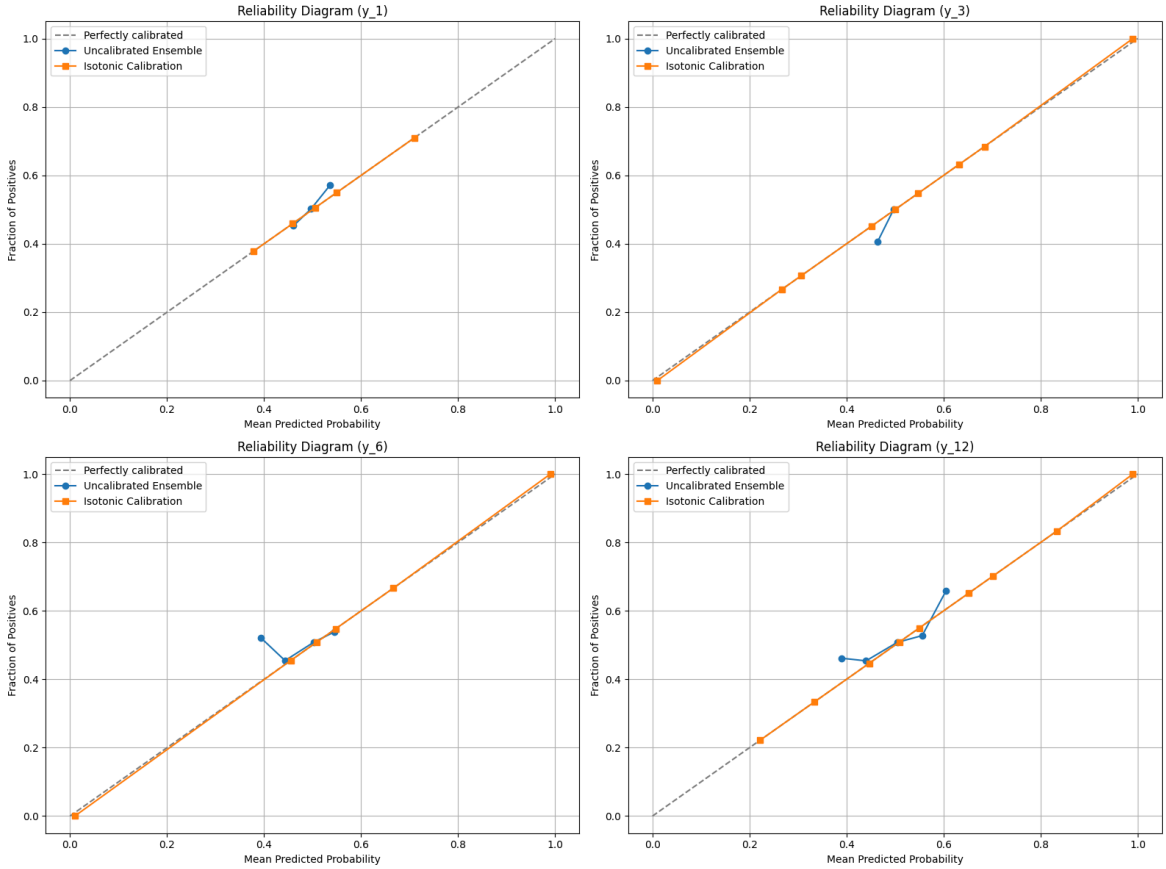
Selective prediction

A futures-style simulator on those gated signals produced strong headline OOS returns (reported +322%, max DD −17.4%, six positive months). That simulator is **not** Polymarket.



Futures-style monthly OOS returns

I also calibrated multi-horizon probability heads for Polymarket-oriented features; isotonic calibration straightened reliability curves, but calibration without edge still loses money at a bad ask.

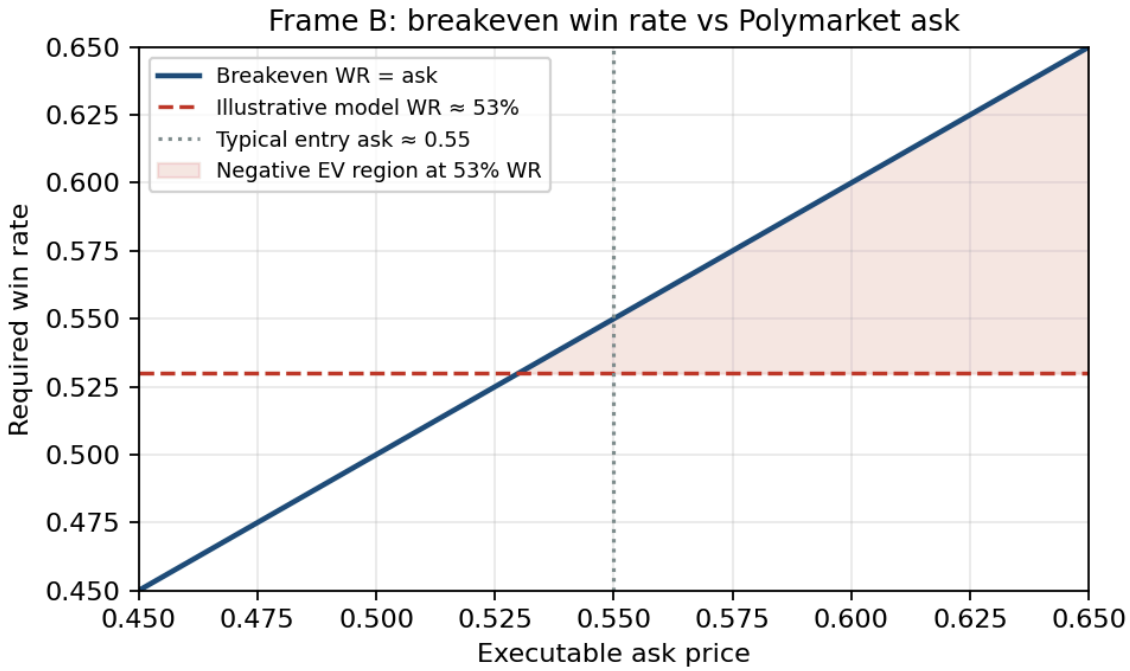


Reliability diagrams

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4. Why Polymarket failed the transfer

4.1 Economics



Breakeven WR vs ask

In live diagnosis the useful operating region looked like ~**53%** hit-rate against asks around **0.54–0.56**. At ask 0.55 that is about **−2% EV per trade**. Breakeven is 55%; a workable edge needs roughly 57–58%+ after selection, or systematically cheaper entries.

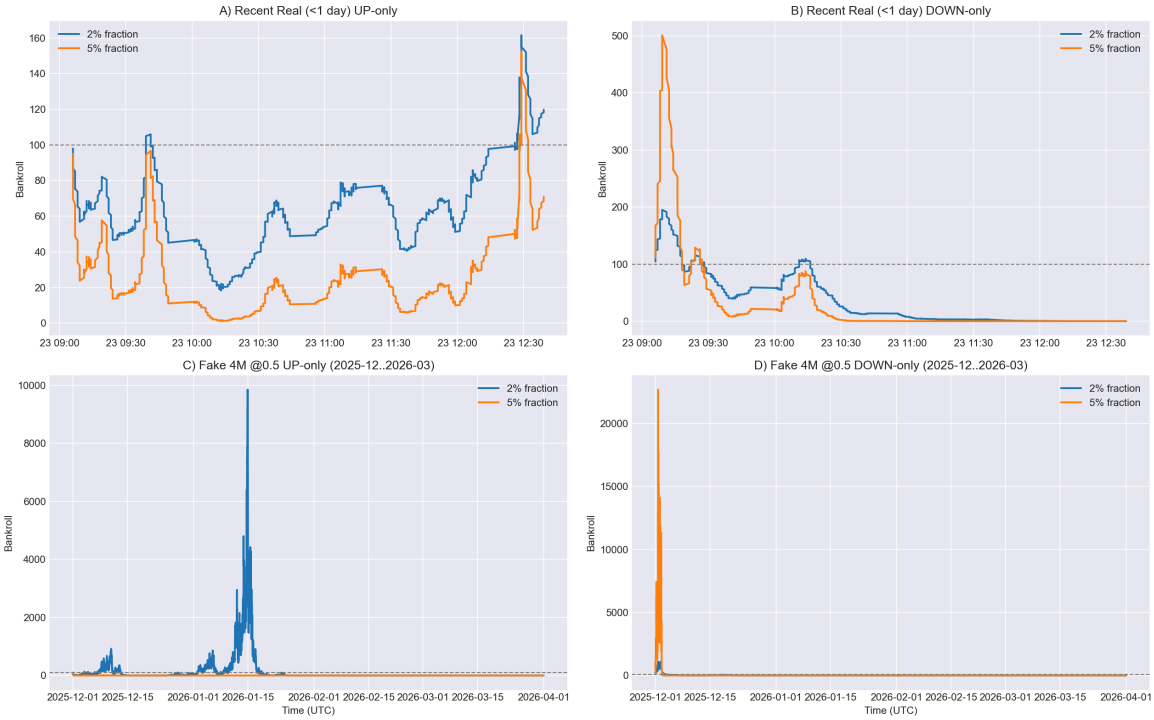
4.2 What broke in practice

See `failure_modes.md`.

- Backtests that assumed mid / even odds overstated edge versus real asks.
- One holdout understated regime variance; walk-forward was stricter.
- Trading every bar diluted rare high-confidence regimes.
- Production plumbing (bot, dry-run/live, redeemer) worked; the **signal** did not clear venue costs.
- Early live ~$200 on a small bankroll is an anecdote.

4.3 What the “explosive bankroll” charts are (and are not)

The figure below is **not** the proof that "Up-only / Down-only with bankroll ramping beats the model." It is a cautionary plot from the lab: **proxy / even-odds bankroll paths** can explode and then vanish, while short **real** windows look unstable. That is a reason to distrust fantasy bankroll curves — not a claim that directional one-sided ramping is the alpha.



Proxy bankroll paths vs short real windows

Actual rejection rule used here: EV at executable ask, plus live/shadow reconciliation — not "compare to Up-only Kelly rocket."

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5. Pipeline

```
Binance OHLCV → features → direction models
        ↓
walk-forward + confidence / regime gates
        ↓
map to Polymarket (side, size, EV vs ask)
        ↓
shadow / dry-run → tiny live
        ↓
truth recompute → keep, gate, or kill
```

Code: `research/features/`, `research/eval/`. Production shape: `production_appendix.md`. Process notes: `decision_log.md`.

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6. Conclusion

Candle models: some short-horizon structure; gating helps; futures-style sim looked strong.
Polymarket: under real asks, the shipped systems did not show a validated edge.

The valuable outcome is the loop that can **kill** a strategy for the right reason.

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7. If continued

1. Walk-forward without tuning gates on the same OOS slice.
2. Reconstruct historical asks for every candidate entry.
3. Optimize calibration / Brier, not accuracy theater.
4. Put microstructure (spread, imbalance, time-in-window) first.

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8. Limits

Curated public extract. Figures mix reported lab tables with process plots. No wallet keys or redeemer secrets.